TELIPORT
Script Your Future
Season 4

VOLT-FORGE

Fewer cells. Clearer trade-offs.

CASE STUDY 2

TATA
TATA ELXSI

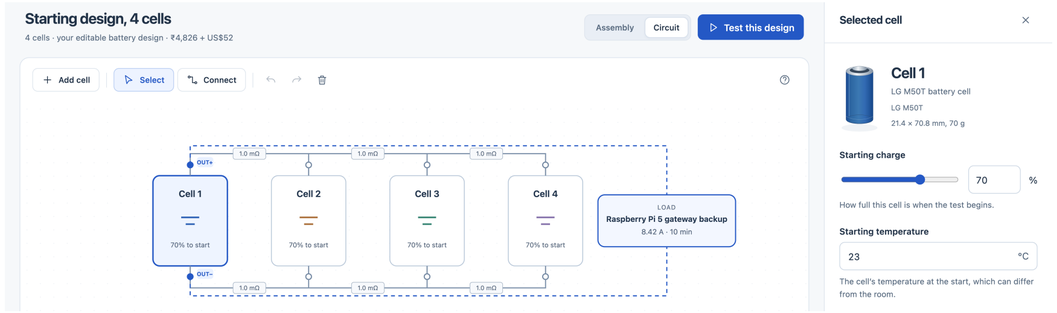
Two cells are a simulation candidate.

Smallest qualifying count among the **1-4-cell parallel arrangements tested** for this backup job.
Half the four-cell baseline's cell cost and mass, with less current and voltage margin.

COMMON BASE CASE Pi 5 backup | LG M50T cells | 25 W supply-rating scenario | 10 min | 23 °C | 70% starting charge

90% converter assumed | 3.3 V floor | 8.42 A constant-current proxy | actual Pi draw not measured

BUILD, TEST, UNDERSTAND



Original V4 editor view; illustrative circuit, not the study table.

For students and lab instructors: edit a circuit, see how current and heat change, then compare alternatives. The search rebuilds and tests each cell count.

BASELINE + TWO OPTIONS

Parallel design	Required cases pass	Worst cell current	Worst cell temperature	Cell mass	Listed cell cost
4 cells | baseline	51/51	3.34 A	26.1 °C	280 g	₹4,824
3 cells	42/42	3.86 A	27.7 °C	210 g	₹3,618
2 cells	**33/33**	**5.49 A**	**31.8 °C**	**140 g**	**₹2,412**

Same-end outputs and declared study links throughout. Cases include mismatch, thermal and converter sensitivities; counts vary with cell positions. All temperatures are simulated. Cost and mass cover cells only.

EVIDENCE

15/15 numerical checks passed

PyBaMM / analytic verification.
All 12 comparison cases missed the voltage and temperature bands. Measured accuracy is not established.

LIMITS THAT MATTER

Conditional, not build-ready

Two cells have no spare. Board battery-current limits remain unverified. Virtual cutoffs are modeled; physical protection has not been demonstrated.

USER + BUYER HYPOTHESIS

Start with teaching labs

Free student use; paid lab licences are an untested pricing hypothesis. No observed user trials yet. Next user test: complete and explain a comparison.

NEXT Measure actual load and converter behavior; bench-check a candidate with supervised cutoffs; observe student tasks.